

Vocalyze — Assistive Digital Governance Platform

AI-Powered Voice Navigation & Inclusive Public Service Delivery

Project Overview & Objective

- **Universal Inclusion:** Designed to empower visually impaired, elderly, and multilingual citizens to independently access and apply for essential government services completely hands-free.
- **Barrier Elimination:** Replaces complex visual web portals with conversational voice prompts and native language interactions.

Key Architectural Capabilities

- **Spoken Voice Assistant:** Automatically greets citizens aloud, announces available public schemes, and navigates portals via natural voice.
- **Interactive Form Engine:** Step-by-step audio narration with smart silence debouncing (1.8s) for accurate multi-digit number capture.
- **Zero-Latency Voice Confirmation:** Rapid verification loops (*Yes/No*) with live on-screen transcripts and spoken correction handling.
- **AI Document OCR:** Auto-extracts identity details from physical documents using Gemini Vision AI and Tesseract.js.
- **Trilingual Support:** Native end-to-end voice and text in English, Hindi (■■■■■), and Kannada (■■■■■).

Technology Stack & Social Impact

- **Frontend:** React 18, Vite, Web Speech API (Synthesis & Recognition), Tesseract.js OCR, WCAG AAA accessible design.
- **Backend & AI:** Node.js, Express, Google Gemini 1.5 Flash API, Firebase Firestore Database & Authentication.
- **Impact:** Ensures equal digital rights, eliminates literacy barriers, and enables direct citizen empowerment.